Rodinia connections between Australia and Laurentia: no SWEAT, no AUSWUS?

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ABSTRACT

Although geological comparisons between Australia and North America have provided a basis for various Neoproterozoic Rodinia reconstructions, quantitative support from precisely dated palaeomagnetic poles has so far been lacking. We report U–Pb ages and palaeomagnetic results for two suites of mafic sills within the intracratonic Bangemall Basin of Western Australia, one of which is dated at 1070 ± 6 Ma and carries a high-stability palaeomagnetic remanence. Comparison of the Bangemall palaeopole with Laurentian data suggests that previous reconstructions of eastern Australia against either western Canada (SWEAT) or the western United States (AUSWUS) are not viable at 1070 Ma. This implies that the Pacific Ocean did not form by separation of Australia–Antarctica from Laurentia, and that up to 10 000 km of late Neoproterozoic passive margins need to be matched with other continental blocks within any proposed Rodinia supercontinent. Our results permit a reconstruction (AUSMEX) that closely aligns late Mesoproterozoic orogenic belts in north-east Australia and southernmost Laurentia.

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Introduction

It has been proposed that most of Earth's continental crust amalgamated during the late Mesoproterozoic (1.3–1.0 Ga) to form the supercontinent Rodinia (Hoffman, 1991). Knowledge of the configuration of the supercontinent is essential to understanding its amalgamation, and its late Neoproterozoic (0.8–0.55 Ga) breakup, which has been linked to extreme environmental and biogeochemical fluctuations and the explosive evolution of metazoan life (Valentine and Moores, 1970; Hoffman *et al*., 1998; Karlstrom *et al*., 2000). However, little consensus has been reached regarding the relative positions of Rodinia's constituent fragments. Most reconstructions place Australia, together with East Antarctica and India, adjacent to either western Canada (the SWEAT hypothesis, e.g. Moores, 1991; Dalziel, 1991; Hoffman, 1991; Powell *et al*., 1993) or the western United States (the AUSWUS hypothesis; Brookfield, 1993; Karlstrom *et al*., 1999; Burrett and Berry, 2000). Although these models are very different (Fig. 1), each is based on matching geological and tectonic features and age provinces (so-called 'piercing points'). The available palaeomagnetic data are presently inadequate to discriminate between the SWEAT and AUSWUS alternatives. In this report we describe an integrated geochronological and palaeomagnetic study of Mesoproterozoic dolerite (diabase) sills in the western Bangemall Basin of Western Australia. The results yield a precisely dated palaeopole, BBS, at 1070 ± 6 Ma, that is the most reliable of this age for Australia. We compare the BBS pole with late Mesoproterozoic poles for Laurentia and explore the implications for reconstructions between these two continents.

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*Details are provided in the accompanying Supplementary Material.

Regional geology

The Mesoproterozoic Bangemall Basin (Fig. 2a) contains more than 6 km of unmetamorphosed, fine-grained carbonate and siliciclastic marine sedimentary rocks known as the Bangemall Supergroup (Muhling and Brakel, 1985; Martin *et al*., 1999a). The basin developed on the site of the 1.83–1.78 Ga Capricorn Orogen, which formed during collision of the Pilbara and Yilgarn Cratons (Tyler and Thorne, 1990; Sheppard and Occhipinti, 2000). The Bangemall Supergroup consists of the lower Edmund Group and overlying Collier Group (Martin *et al*., 1999a). The Edmund Group contains ~ 4 km of stromatolitic dolomite and fine-grained clastic sediments, and unconformably overlies deformed Palaeoproterozoic rocks of the Ashburton and Bresnahan Basins in the north, and the igneous and metamorphic Gascoyne Complex in the south-west. The unconformably overlying Collier Group contains ~ 3 km of siltstone and sandstone, and, in the west, occupies a regional synclinorium that extends along the basin axis. Extensive quartz dolerite sills, classified geochemically as high-Ti continental tholeiites (Muhling and Brakel, 1985), occur throughout the Bangemall succession and are mainly concordant with bedding, although locally they transgress the stratigraphy. Sills are typically > 100 m thick, mainly medium-grained, with locally exposed chilled margins and coarse-grained phases.

The northern margin of the Bangemall Basin is relatively undeformed where it overlies the Ashburton Basin (Fig. 2a), which acted as a stable shelf during deposition. Following intrusion of dolerite sills, the southern parts of the basin were compressed northwards against the Ashburton shelf, resulting in an arcuate region of elongate, tight to open folds known as the Edmund Fold Belt (Muhling and Brakel, 1985). Folding may have been related to 1090–1060 Ma tectonothermal events (Bruguier *et al*., 1999) in the adjacent Darling Mobile

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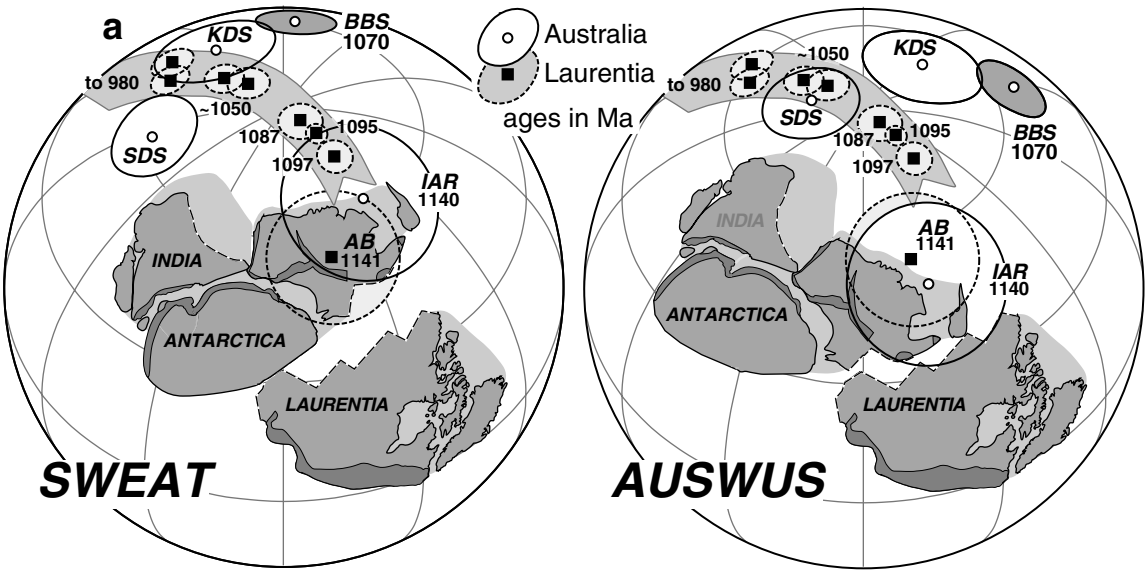


Fig. 1 Early to mid-Neoproterozoic reconstructions of East Gondwanaland (Australia + East Antarctica + India) and Laurentia, according to (a) the SWEAT (*south-west* US – *E*ast An*t*arctica) and (b) AUSWUS (*Aus*tralia – *w*estern *US*) hypotheses. The 1070 ± 6 Ma BBS pole (shaded) obtained in this study is incompatible with Laurentian data in any reconstruction similar to SWEAT or AUSWUS. 'Grenville-age' mobile belts are shaded dark grey. Palaeopoles defining the Laurentian apparent polar wander (APW) path are described in the Supplementary material; also shown is the pole (AB) from the 1141 Ma Abitibi dykes (Ernst and Buchan, 1993). Australian palaeopoles include those for the 1140 Ma Lakeview dolerite, IAR (Tanaka and Idnurm, 1994; J. Claouè-Long, pers. comm., 2001), Stuart dykes, SDS (Idnurm and Giddings, 1988), and Kulgera sills, KDS (Camacho *et al.*, 1991). All palaeopoles (additional details are provided in Fig. 5 and the Supplementary material) are shown with α_{95} confidence circles; arbitrary lines of longitude are 30° apart.

Belt (Fig. 2a), and certainly occurred prior to intrusion of N- to NE-trending dolerite dykes of the 755 Ma Mundine Well swarm (Wingate and Giddings, 2000), which are essentially undeformed and cut across all older rocks and fabrics.

Geochronology

SHRIMP zircon ages of 1679 ± 6 and 1619 ± 15 Ma for underlying intrusions provide an older limit for Bangemall sedimentation (Pearson *et al.*, 1995; Nelson, 1998). SHRIMP U–Pb analyses of xenocrystic zircons from altered rhyolite near the base of the Edmund Group yielded a maximum age for eruption of 1638 ± 14 Ma (Nelson, 1995). Several K–Ar and Rb–Sr studies of dolerites and baked sedimentary rocks produced ages between 1050 and 1075 Ma (Compston and Arriens, 1968; Gee *et al.*, 1976; Goode and Hall, 1981). Our SHRIMP U–Pb results* indicate that dolerite sills were emplaced during two distinct events. For three samples (sites 1, 7 and 10; Fig. 2b), all baddeleyite and zircon $^{207}Pb/^{206}Pb$ ratios agree to within analytical precision and yield statistically identical ages of 1071 ± 8, 1067 ± 14 and 1068 ± 22 Ma (Fig. 3). The results are combined to yield a mean age of 1070 ± 6 Ma (95% confidence interval), which we regard as the time of crystallization of the younger sill suite. Zircon and baddeleyite from two sites (11 and 21) provide a mean age for the older sills of 1465 ± 3 Ma.

Palaeomagnetism

Samples were collected from dolerite sills and sedimentary rocks at 25 sites throughout the western Bangemall Basin (Fig. 2b). After removal of low-coercivity overprints by alternating field (AF) demagnetization to 10 or 20 mT, two main types of magnetic behaviour were observed. The majority of samples yielded an inconsistently directed remanence of low thermal stability (referred to as type L), which we interpret as a chemical remanent magnetization (CRM) carried mainly by maghemite*. A consistently directed magnetization (referred to as type A) was isolated in 79 samples from 15 sites (Table 1), including the three dated at *c*. 1070 Ma, and is the only stable remanence present at five sites. Unblocking temperatures between 500 °C and 580 °C show the remanence to be single component and that relatively pure magnetite is the dominant carrier (Fig. 4a). Most specimens are stable to AF treatment of 100–160 mT, and decay curves indicate variable proportions of MD and SD grains. Shale at five sites yields A magnetizations with directions similar to that of adjacent dolerite. Site mean directions converge after correction for bedding tilt, and the concentration parameter, k, increases from 6 to 30 (Table 1). Corrected directions are NNW with moderate downwards inclination, except at site 25, where the direction is SSE and upward (Fig. 4b). The mean direction, after tectonic correction (and inversion of data from site 25), is D, $I =$ 339.3°, 46.5° ($\alpha_{95} = 8.4°$, $N = 11$ sites).

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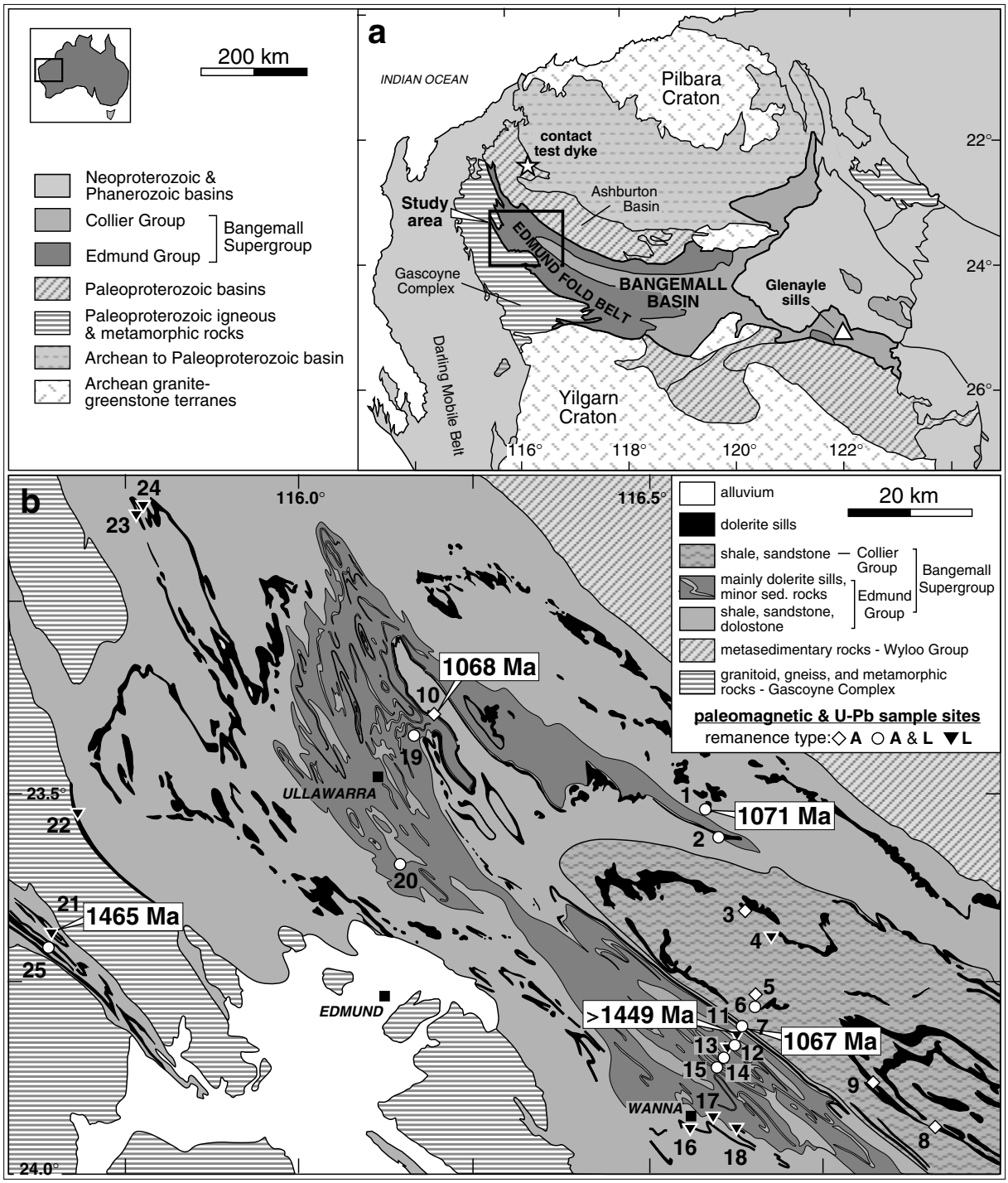


Fig. 2 (a) Location and geological setting of the Bangemall Basin. The triangle and star indicate the location of dolerite sills in the eastern Bangemall Basin, and of a dolerite dyke, both of which yield palaeomagnetic directions similar to those obtained in this study from 1070 Ma sills. (b) Simplified geological map of the western Bangemall Supergroup (Daniels, 1969; Copp *et al.*, 1999; Martin *et al.*, 1999b), showing the locations of samples collected for palaeomagnetism and U–Pb geochronology (ages in Ma). Symbols indicate the palaeomagnetic remanence type present at each site.

Several lines of evidence indicate a primary origin for the A magnetization. (1) Low within-site dispersion is typical of primary thermoremanent magnetizations (TRM) in rapidly cooled intrusions. (2) Positive fold tests* show that the A magnetization was acquired prior to folding, which may have occurred soon after sill

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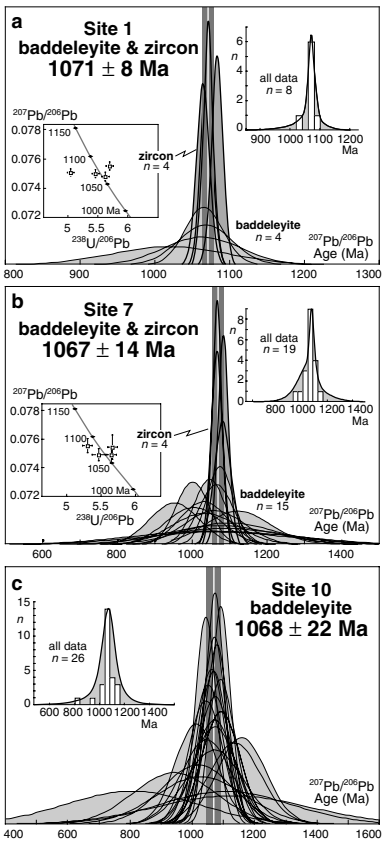


Fig. 3 (a–c) Normalized probability density curves for individual SHRIMP $^{207}Pb/^{206}Pb$ ages measured in baddeleyite and zircon (bold curves) from sites 1, 7 and 10. Dark grey bands indicate the weighted mean age and 95% confidence intervals. Left-hand insets in (a) and (b) show U–Pb analytical data for zircons; other insets superimpose cumulative probability density curves and histograms for all data. *n*, number of analyses.

intrusion at 1070 Ma. (3) Although SD magnetite grains in some sill samples require heating close to 580 °C to unblock their magnetization, previous K–Ar and Rb–Sr studies yielded ages between 1050 and 1075 Ma, and there is no evidence for a thermal event between 1070 and 755 Ma that could cause a remagnetization. (4) The presence of polarity reversals between, but not within, intrusions is supportive of a primary remanence. (5) Sedimentary rocks in baked contacts appear to be overprinted by the A magnetization. Although no stable remanence was isolated in unbaked rocks, an undated NNE-trending dyke (star, Figs 2a and 4b) carrying the Bangemall A direction yields a positive baked-contact test* with its host rock, the 2.45 Ga Woongarra Rhyolite. This suggests that the dyke is similar in age to the sills (possibly comagmatic), and that the A magnetization in the sills is also original.

We conclude that the A component is a primary TRM acquired during sill emplacement at 1070 Ma. Directions of opposite polarity at site 25 imply that the intrusive event spanned at least one reversal of the Earth's magnetic field, and that, collectively, the A magnetizations adequately average palaeosecular variation. Similar palaeomagnetic directions obtained from undeformed sills in the Glenayle area*, in the eastern Bangemall Basin (triangle, Figs 2a and 4b), together with the lack of significant deformation of the Mundine Well dyke swarm, indicate that the Bangemall Basin has undergone no internal vertical-axis rotation since 1070 Ma. The palaeomagnetic pole, BBS, lies at 33.8°N, 95.0°E ($\alpha_{95} = 8.3°$).

Comparison with previous results

Late Mesoproterozoic palaeopoles for Australia were obtained previously from the Stuart dykes and Kulgera sills in central Australia (Idnurm and Giddings, 1988; Camacho *et al.*, 1991). The Stuart dykes yielded Sm–Nd and Rb–Sr isochron ages of 1076 ± 33 and 897 ± 9 Ma, respectively (Black *et al.*, 1980; Zhao and McCulloch, 1993). The predominantly N-trending dolerite dykes, now locally sheared and altered, were intruded into Palaeoproterozoic basement granitoids of the southern Arunta Block, which was deformed strongly and uplifted during the Carboniferous Alice Springs orogeny (Collins and Shaw, 1995). Reliability of the preliminary SDS palaeopole is difficult to assess because no data or analytical details have been published. The shallowly S- to SE-dipping Kulgera sills yielded Sm–Nd and Rb–Sr isochron ages of 1090 ± 32 and 1054 ± 14 Ma, respectively (Camacho *et al.*, 1991; Zhao and McCulloch, 1993). Kulgera sills were intruded into Mesoproterozoic gneisses of the eastern Musgrave Block, which subsequently experienced late Neoproterozoic (Petermann Ranges) and Carboniferous (Alice Springs) tectonothermal events. Reliable constraints on palaeohorizontal are not available for the Kulgera or Stuart intrusions (no tectonic corrections were applied), and both suites are located in crustal blocks that were deformed and probably re-orientated after dyke emplacement. Although isotopic data suggest that the Stuart and Kulgera intrusions are similar in age, their palaeopoles are significantly different (Fig. 1). The BBS palaeopole does not agree with the KDS or SDS poles, but is more reliable than either. The BBS pole is inferred to be primary, is dated precisely, and structural control is well-defined in adjacent sedimentary rocks. The BBS pole achieves a perfect score of Q = 7 in the reliability scheme of Van der Voo (1990).

Implications for Rodinia reconstructions

The SWEAT reconstruction (Fig. 1a) was constrained by optimizing the fit between Australian and Laurentian poles at ~ 1070 and at 700–750 Ma (Powell *et al.*, 1993). However, the previous 1070 Ma poles for Australia are unreliable, as described above, and the supposedly 700–750 Ma YB dykes pole for Australia (Giddings, 1976) may represent a younger (possibly Mesozoic) overprint (Halls and Wingate, 2001). Palaeomagnetic support for the AUSWUS reconstruction (Fig. 1b) was based on matching Australian and Laurentian poles between ~ 1.75 and 0.75 Ga (Karlstrom *et al.*, 1999; Burrett and Berry, 2000). However, most Mesoproterozoic data for Australia are of low reliability, or are dated inadequately. A precisely dated primary pole for the Mundine Well dykes of Australia permits neither SWEAT nor AUSWUS at 755 Ma (Wingate and Giddings, 2000), although this result by itself allows either reconstruction to have been valid for earlier times. Our new BBS palaeopole permits a direct test of proposed fits at 1070 Ma, prior to any plausible age for Rodinia's fragmentation (Hoffman, 1991). Although there is no well-dated pole for Laurentia at 1070 Ma, the trend of the Laurentian APW path* between 1100 and ~ 1020 Ma is well defined. The BBS pole is separated by ~ 30° from the Laurentian path in the SWEAT fit

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Table 1 Site mean directions and virtual geomagnetic poles (VGPs) for sites exhibiting A-type magnetization

	Location		Bedding		Direction (*in situ*)				Direction (rotated)		VGP	
Site	Lat. (°S)	Long. (°E)	(dip/dip azimuth)	*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	*D′* (°)	*I′* (°)	Lat. (°N)	Long. (°E)
1	23.522	116.580	13/088	9(3)	344.9	48.4	75	5.6	359.8	48.7	**36.8**	116.4
2*	23.557	116.597	35/213	(2)	017.6	22.3	151	14.4	007.4	54.9	30.6	123.6
3	23.653	116.633	04/353	5	336.4	37.7	108	6.6	337.2	33.9	42.3	86.9
5	23.765	116.650	01/134	(5)	328.5	50.5	610	2.8	328.5	51.5	26.5	87.0
6	23.783	116.652	02/320	8(3)	337.4	37.1	41	8.2	337.0	35.2	41.3	87.3
7	23.807	116.630	60/041	6(2)	267.9	49.3	37	10.1	352.4	50.6	34.4	108.8
8	23.938	116.910	29/029	6	312.1	57.7	257	3.8	343.7	42.9	38.7	97.9
9	23.875	116.820	19/233	6	344.7	40.9	97	6.2	327.2	45.2	30.3	82.7
10	23.401	116.192	43/028	6	268.1	65.3	575	2.6	350.2	53.8	31.5	106.7
12	23.822	116.626	80/034	5	244.8	22.6	195	4.9	331.0	58.0	21.9	92.6
14*	23.838	116.613	43/072	2	324.1	52.3	1436	4.7	016.6	45.0	37.1	135.3
15	23.855	116.606	74/205	4	002.8	5.1	20	18.1	319.9	65.2	10.1	90.2
19*	23.426	116.172	07/178	2	346.3	47.0	151	14.4	344.8	53.3	30.9	101.5
20*	23.589	116.145	65/055	2	268.6	46.5	183	13.1	012.5	55.7	29.0	127.7
25	23.708	115.633	79/038	11	111.3	−37.5	40	6.9	164.1	−19.9	52.6	89.3
dyke	22.197	116.290	–	9	341.9	60.0	159	4.1	–	–	24.7	101.3
Mean direction (*in situ*):				11 sites	316.7	47.9	6	19.8	Palaeopole:		33.8	95.0
Mean direction (rotated):				11 sites	339.9	46.5	30	8.4	$n = 11$ sites, $K = 32$, $A_{95} = 8.3°$			

Results from four sites (marked with asterisks) that each yielded two data points, although consistent with the remaining data, are considered less reliable and were not included in calculating mean results. Results listed for the dyke (including its baked contact) were also not included in calculations. *n*, number of samples given unit weight in calculation of mean (number of sedimentary rock samples included in parentheses); *D*, declination (E of N); *I*, inclination (positive downwards); *k* (*K*), Fisher's precision parameter; α_{95}, (A_{95}), circle of 95% confidence about mean direction (pole). VGPs are calculated from rotated (bedding-corrected) directions.

and by at least 40° in the AUSWUS fit (Fig. 1). Moreover, a fit similar to SWEAT or AUSWUS cannot be achieved by matching the BBS pole with any part of the Laurentian path shown in Fig. 1, indicating that neither reconstruction is viable at 1070 Ma.

To explore possible reconstructions between Australia and Laurentia, we approximate a pole position for Laurentia at 1070 Ma by assuming constant APW between the 1087 and ~ 1050 Ma poles. Superimposing the BBS palaeopole and this interpolated pole position for Laurentia at the projection axis (Fig. 5) places the continents at their correct orientations and palaeolatitudes at 1070 Ma. Australia is situated at lower palaeolatitudes than is permitted by the SWEAT or AUSWUS models, placing the Cape River Province of north-east Australia at a similar latitude to the south-west end of the ~ 1250–980 Ma Grenville Province of Laurentia (Rivers, 1997). High-grade metamorphic and magmatic rocks in the Cape River Province contain 1240, 1145 and 1105 Ma zircon age components, and may correlate with 'Grenville-age' rocks in the Musgrave and Albany–Fraser orogens (Blewett *et al.*, 1998). The Grenville Province may therefore have continued through Australia. The time at which Australia and Laurentia might have come together is unknown, but can be investigated by comparing older palaeopoles from each block. The tight reconstruction in Fig. 5 permits the α_{95} confidence circles of the 1140 Ma IAR and AB poles to overlap, although these two poles are insufficiently precise for a rigorous test.

Owing to the lack of palaeolongitude control, it is entirely possible that Australia and Laurentia were not joined at 1070 Ma. In addition, because final 'Grenvillian' assembly of Rodinia could conceivably post-date 1070 Ma, there remains the possibility that fits similar to SWEAT or AUSWUS could have been achieved through a post-1070 Ma collision between a unified Australian–Mawson craton and some part of the proto-Cordilleran Laurentian margin. In this case an intermediary craton would appear to be required to bear the main record of such a collision (e.g. South China; Li *et al.*, 2001), evidence for which is largely lacking in eastern Australia and western North America. Moreover, similarities among Palaeoproterozoic and early Mesoproterozoic rocks in Australia and North America would be fortuitous, removing some of the very foundations for the SWEAT and AUSWUS models.

The provocative fit suggested in Fig. 5, referred to here as AUSMEX (Australia – Mexico), requires further testing by comparing additional Proterozoic palaeopoles of precisely the same age from Australia and Laurentia, with subsequent reconstructions to be elaborated using geological and other constraints. The most compelling geological arguments used to generate the SWEAT and AUSWUS hypotheses, including correlation of Mesoproterozoic orogenic belts, Palaeo- and Mesoproterozoic isotopic age provinces, and Neoproterozoic rift – passive margin sedimentary successions, remain robust in the AUSMEX reconstruction. The SWEAT and AUSWUS models implied that Neoproterozoic separation of Australia–Antarctica from Laurentia led to opening of the Pacific Ocean. The results of this study, however, suggest

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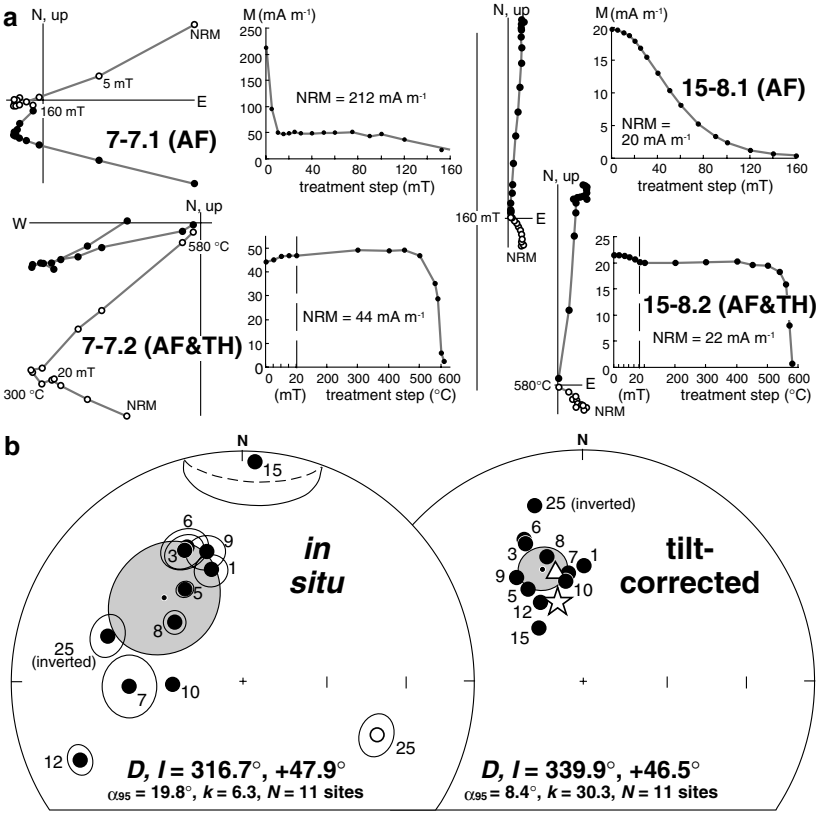


Fig. 4 (a) Examples of alternating field (AF) and thermal (TH) demagnetization of A-type remanence in two specimens of a single core from each of sites 7 and 15. Orthogonal projections show trajectories of vector endpoints during progressive demagnetization (open/closed symbols represent vertical/horizontal plane). Demagnetization curves show changes in intensity during treatment. Reference frame is present horizontal. (b) Component A site mean directions in geographical (*in situ*) and stratigraphic (tilt-corrected) coordinates. Open/closed symbols in lower-hemisphere equal-angle stereographic projections indicate upward/downward pointing directions. The overall mean direction in each case is shown with a shaded α_{95} circle. The mean direction for site 25 was inverted through the origin prior to statistical calculations. The star and triangle show mean palaeomagnetic directions obtained for a NNE-trending dyke that yields a positive baked-contact test, and for nine sites in undeformed dolerite sills in the eastern Bangemall Basin, respectively (described in the Supplementary material). *D*, declination (E of N); *I*, inclination (positive downwards); α_{95}, semi-angle of cone of 95% confidence about mean direction; *k*, Fisher's precision parameter.

that western Laurentia was not the conjugate margin to eastern Australia–Antarctica. The origin of the Pacific Ocean is therefore undetermined, and up to 10 000 km of late Neoproterozoic passive margins in eastern Australia and western Laurentia need to be matched with other continental blocks within any proposed Rodinia supercontinent.

Supplementary material

The following material is available from http://blackwell-science.com/products/journals/suppmat/TER/TER401/TER401sm.html

Table S1 Ion microprobe data for baddeleyite and zircon from site 1.

Table S2 Ion microprobe analytical data for baddeleyite and zircon from site 7.

Table S3 Ion microprobe data for baddeleyite from site 10.

Table S4 Selected late Mesoproterozoic palaeopoles for Laurentia.

Fig. S1 Examples of AF and thermal demagnetization of L-type remanence in two specimens of a single core from each of sites 15 (a) and 25 (b). Orthogonal projections show trajectories of vector endpoints during progressive demagnetization (open/closed symbols represent vertical/horizontal plane). Open/closed symbols in lower-hemisphere equal-area stereographic projections indicate upward/downward pointing directions. Reference frame is present horizontal. Demagnetization curves show changes in magnetization intensity during treatment.

Fig. S2. Directions and thermal demagnetization for A- and L-type magnetizations at site 15 (a) and L-type magnetizations at site 21 (b). Demagnetization curves are normalized to values at 100 °C. The mean direction and α_{95} confidence circle are shown for four A-type directions at site 15. Other notes as in Fig. S1.

Fig. S3. (a) Outcrop sketch for the baked-contact test between a dolerite dyke and Woongarra Rhyolite. (b) Palaeomagnetic directions; α_{95} confidence circles are shown around the mean for each group. (c) Examples of thermal demagnetization of four specimens (distance from dyke contact shown in parentheses). Other notes as in Fig. S1.

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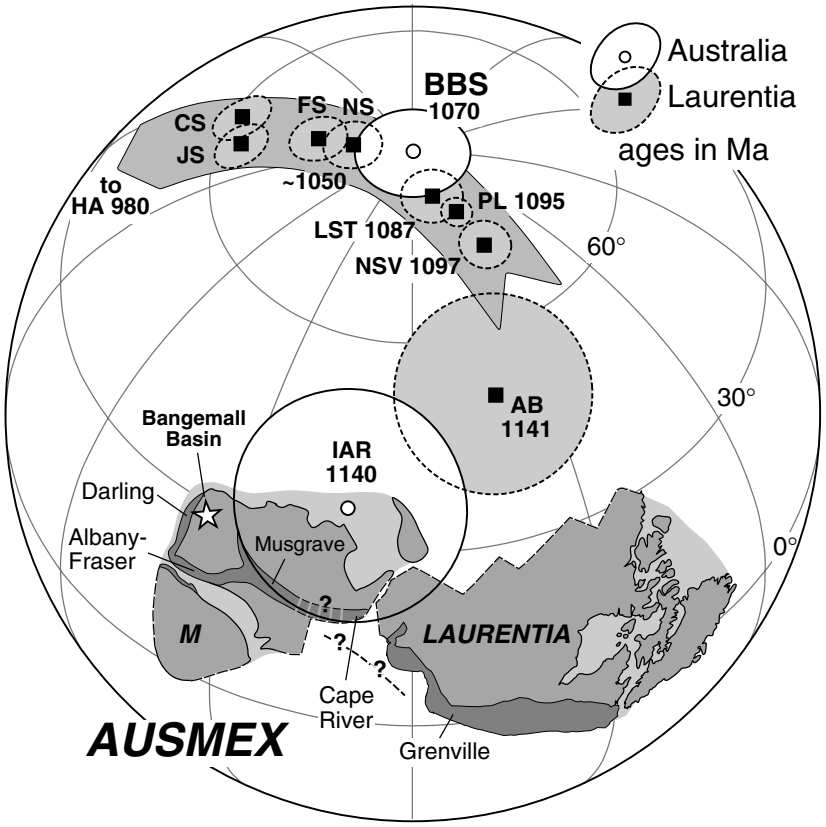


Fig. 5 A possible reconstruction (AUSMEX) between Australia and Laurentia at 1070 Ma that places the Grenville and Cape River provinces at similar palaeolatitudes. Note that palaeolongitudes are not constrained when comparing individual pole positions. M is the Mawson block of East Antarctica according to Fitzsimons (2000). Orogenic belts discussed in the text are labelled. Rotation parameters (clockwise positive): Australia to absolute frame, 41.49°N, 34.51°E, 79.53°; Laurentia to Australia, 54.52°N, 149.86°E, – 120.80°. Other notes as in Fig. 1.

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